FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – III

JEE (Main)-2022

TEST DATE: 01-06-2022

ANSWERS, HINTS & SOLUTIONS

Physics PART – A

SECTION – A

1. C

Sol. The force acting on the particle = $\frac{mdv}{dt}$

Power of the force = $\left(\frac{mdv}{dt}\right)$ v = k (constant)

$$m\,\frac{v^2}{2} = kt + c \qquad ...(1)$$

at $t = 0, v = u \quad \therefore c = \frac{mu^2}{2}$

Now from (1),

$$m\frac{v^2}{2} = kt + \frac{mu^2}{2}$$

$$\frac{1}{2}m\,(v^2 - u^2) = kt \qquad ...(2)$$

Again $\frac{mdv}{dt}v = k$

$$m.v\,\frac{dv}{dx}v = k$$

$$\int_u^v mv^2 dv = \int_0^x k\,dx$$

Intergrating, $\frac{1}{3}m\,(v^3 - u^3) = kx$ ...(3)

From eqn (2) and (3),

$t = \frac{3}{2}\left(\frac{v^2 - u^2}{v^3 - u^3}\right)x$

2. A

Sol. From ohm's law

Electric field α current density

3. D

Sol. $e = \int_{-\ell/2}^{\ell/2} B\omega r dr = 0$

4. A

5. B

6. B

Sol. When current is maximum $= \frac{di}{dt} = 0$

7. D

8. A

Sol. Saturation for A and B are different hence intensities are different. Stopping potentials for A and B are equal hence frequencies are equal.

9. B

Sol. Characteristic wavelengths shift of left if z is increased hence B is incorrect

10. D

11. B

Sol. $\rho_1 g h_1^2 = \rho_2 g h_2$

$\frac{h_1}{h_2} = \frac{\rho_2}{\rho_1}$

12. B

13. A

Sol. Process A $\rightarrow$ B is isothermal

Process B $\rightarrow$ C isochoric

Process C $\rightarrow$ A is isobaric

14. C

Sol. The process described is isochoric

Therefore molar heat capacity $= C_v = 2.5R$

15. A

16. B

Sol. Capacitors are in series therefore

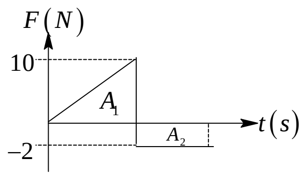
$$\frac{C_1}{C_2} = \frac{V_2}{V_1} = \frac{2}{3}$$

17. B

18. D

19. C

Sol.



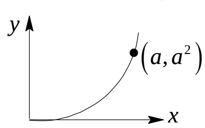
Change in momentum = impulse = Net area under F-t graph

$$= \left(\frac{1}{2}\times 10\times 10\right) - (10\times 2) = 30\,kg\,ms^{-1}$$

20. A

Sol. $\overline{F} = xy^2\hat{i} + yx^2\hat{j}$, and displacement

$d\overline{r} = dx\hat{i} + dy\,\hat{j}$



$$W = \int_{Path} \overline{F}.d\overline{r} = \int_{Path} \left(xy^2\hat{i} + yx^2\hat{j}\right).\left(dx\hat{i} + dy\,\hat{j}\right)$$

$$= \int_{Path} \left(xy^2dx + yx^2dy\right) = \int_{x=0}^{a} xy^2dx + \int_{y=0}^{y=a^2} yx^2dy$$

$$W = \int_{x=0}^{a} x^5dx + \int_{y=0}^{a^2} y^2dy = \frac{a^6}{6} + \frac{a^6}{3} = \frac{a^6 + 2a^6}{6} = \frac{a^6}{2}$$

SECTION – B

21. 00000.00

Sol. At $t = 0, x = \dfrac{4}{3}\,and\,v = 0$

At $x = \dfrac{4}{3}m, a = 3\left(\dfrac{4}{3}\right) - 4 = 0$

22. 00004.00

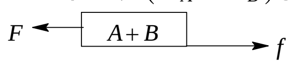
Sol. The distance travelled (here it is displacement) is are under $v - t$ graph

$$= \frac{1}{2}\times 2\times 2 + \frac{1}{2}\times 2\times 2 = 4m$$

23. 00120.00

24. 00003.00

Sol. $F = f = \mu\left(m_A + m_B\right)g = 0.25 \times 12 = 3N$



25. 00000.12

26. 00006.00

27. 00001.00

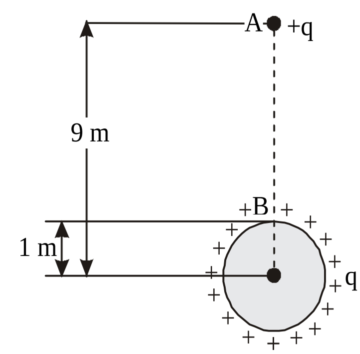
Sol. Use $a = \frac{dV}{ds}$ and find $\frac{dV}{ds}$ using the slope of normal

28. 00060.02

Sol. Since the scale is graduated at 10ºC,
1 cm of the scale at 10ºC = exactly 1 cm
$\therefore$ 1 cm of the scale at 30ºC
= exactly $(1 + 18 \times 10^{-6} \times 20)$cm
$\therefore$ 60 cm of the scale at 30ºC
= exactly $60.00\,(1 + 36 \times 10^{-5})$
= 60.02 cm

29. 00028.00

Sol. Keeping in mind that here both electric and gravitational potential energy are changing and for external point a charged sphere behaves as whole of its charge were concentrated at its centre, applying conservation of energy between initial and final position, we have



$$\frac{1}{4\pi\varepsilon_0}\frac{qq}{9} + mg \times 9 = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{1} + mg \times 1$$

or $q^2 = \frac{80 \times 10^{-3} \times 9.8}{10^9}$

or $q = 28\ \mu C$

30. 06000.00

Sol. The fringe–width β is given by

$\beta = \frac{\lambda D}{2d}$ and $\beta' = \frac{\lambda D'}{2d}$

where λ is the wavelength of light used, D is the distance of the screen from the two slits and 2 d is the separation between two slits.

Now $\beta - \beta' = \frac{\lambda(D - D')}{2d}$

$\Rightarrow \lambda = \frac{(\beta - \beta')2d}{D - D'}$

$\lambda = \frac{(3 \times 10^{-5})(10^{-3})}{5 \times 10^{-2}}$

$= 0.6 \times 10^{-6}$ m = 6000 Å

Chemistry

PART – B

SECTION – A

31. A
32. A
33. C
34. C
35. A
36. B
37. B
38. B
39. D
40. C
41. D
42. B
43. A
44. C
45. A
46. B
47. D
48. A
49. D
50. D

SECTION – B

51. 00019.48
52. 00750.00
53. 00003.00
54. 00009.00
55. 00005.00

56. 00001.65

Sol. Given that $\Delta E - \Delta H = 1200\text{cal}$

Where ΔE is heat of reaction at constant volume and ΔH is heat of reaction at constant pressure.

Also we have $\Delta H = \Delta E + \Delta$nRT

$\therefore nRT = -1200$

Or $\Delta n = \dfrac{-1200}{2\times 300} = -2$

Now $K_P = K_C\ (RT)^{\Delta n}$

$\therefore \dfrac{K_P}{K_C} = \left(0.0821\times 300\right)^{-2} = 1.648\times 10^{-3}$

57. 00003.52

Sol. First dissociation: $H_3X \longrightarrow H_2X^- + H^+$

$\therefore H^+$ ion concentration $= 2\times 10^{-4}$M

Second dissociation: $H_2X^- \rightleftharpoons HX^- + H^+$

$\therefore H^+$ ion concentration = $C\alpha = 2\times 10^{-4}\times 0.5 = 10^{-4}$

$\because$ Third dissociation is negligible

$\therefore \left[H^+\right] = 2\times 10^{-4} + 10^{-4} = 3\times 10^{-4}$

pH=4-log3=3.52

58. 00008.00

Sol. The rate constant for the reaction is $k = k_1 + k_2 = 2.0\times 10^{-5} + 4.0\times 10^{-5} = 6.0\times 10^{-5}s^{-1}$. From first order kinetics, we have $kt = 2.303\log\dfrac{[A]_0}{[A]_t}$

Substituting values, we get

$6\times 10^{-5}\times 3\times 60\times 60 = 2.303\log\dfrac{0.25}{[A]_t} \Rightarrow \dfrac{0.25}{[A]_t} = \text{antilog}\left(0.2814\right) = 1.91$

Or $[A]_t$ =0.13M. Therefore, the amount of $[A]$ decomposed $= [A]_0 - [A]_t$ =0.25-0.13=0.12M

Thus, fraction of C formed is

$\dfrac{k_1}{k_1+k_2}\times [A](\text{decomposed})\times \dfrac{1}{1} = 0.12\times \dfrac{4\times 10^{-5}}{6\times 10^{-5}} = 8\times 10^{-3}\text{M}$

59. 00008.00

Sol.

Complex	**No. of enantiomer pairs**
$\left[M(AB)a_3b\right]$	1
$\left[M(AB)a_2b_2\right]$	2
$\left[M(AB)a_2bc\right]$	5

60. 00006.00

Sol.

$ClF(g)$ Colourless		
	$ClF_3(g)$ Colourless	
$BrCl(g)$ Red-brown	$ClF_3(g)$ Colourless	
$ICl(s)$ Ruby red	$BrF_5(l)$ Colourless	$IF_7(g)$ Colourless
	$IF_5(l)$ Colourless	

Mathematics

PART – C

SECTION – A

61. B

Sol. Given $\frac{2ab}{a+b}:\sqrt{ab}=\left(4r^2-1\right):\left(4r^2+1\right)$

$\Rightarrow \frac{a+b}{2\sqrt{ab}}=\frac{4r^2+1}{4r^2-1}$

$\Rightarrow \frac{\sqrt{a}+\sqrt{b}}{\sqrt{a}-\sqrt{b}}=2r$

$\Rightarrow \frac{2\sqrt{a}}{2\sqrt{b}}=\frac{2r+1}{2r-1}$

$\Rightarrow \sqrt{\frac{a}{b}}=\left(1+\frac{1}{2r}\right)\left(1-\frac{1}{2r}\right)^{-1}$

$\Rightarrow \sqrt{\frac{a}{b}}=1+\frac{1}{r}$ (neglecting $\frac{1}{r^2},\frac{1}{r^3}......$)

62. D

Sol. Given $\sum x\left(b+x\right)\left(c+x\right)\left(d+x\right)=2\left(a+x\right)\left(b+x\right)\left(c+x\right)\left(d+x\right)$

$\Rightarrow 2x^4+\left(a+b+c+d\right)x^3+0.x^2-\left(abc+abd+acd+bcd\right)x-2abcd=0$

This is a fourth degree equation whose two roots are ω and ω^2. Let α and β be the two other roots, then $\left(\alpha+\beta\right)\left(\omega+\omega^2\right)+\alpha\beta+\omega.\omega^2=0$

$\Rightarrow \left(\alpha+\beta\right)\left(-1\right)+\alpha\beta+1=0$

$\Rightarrow \alpha=1\ or\ \beta=1$

Product of roots $=\frac{-2abcd}{2}=1.1.\omega.\omega^2$

$\Rightarrow -abcd=1$

$\Rightarrow abcd=-1$

63. C

Sol. Let $I=\int\frac{dx}{\sin x\left(2+\cos x-2\sin x\right)}$

Let $\tan\frac{x}{2}=t$

$\Rightarrow \frac{1}{2}\sec^2\frac{x}{2}dx=dt$

$\Rightarrow dx=\frac{2dt}{1+t^2}$

$$\therefore\ I = \int \frac{\frac{2dt}{1+t^2}}{\frac{2t}{1+t^2}\left(2+\frac{1-t^2}{1+t^2}-\frac{4t}{1+t^2}\right)}$$

$$= \int \frac{\left(1+t^2\right)dt}{t\left(t-3\right)\left(t-1\right)}$$

$$\therefore\ I = \frac{1}{3}\ell n\left|\tan\frac{x}{2}\right| + \frac{5}{3}\ell n\left|\tan\frac{x}{2}-3\right| - \ell n\left|\tan\frac{x}{2}-1\right| + c$$

64. D

Sol. $\bar{x} = \frac{a+\left(a+d\right)+...........+\left(a+2nd\right)}{2n+1}$

$= a + nd$

$\therefore$ MD from mean $= \frac{1}{2n+1}\sum\left|x_i - \bar{x}\right|$

$$= \frac{1}{2n+1}2\left|d\right|\left(1+2+3+...........+n\right)$$

$$= \frac{n\left(n+1\right)\left|d\right|}{2n+1}$$

65. C

Sol. Since, $\vec{a} = \vec{x} - \vec{y}$ & $\vec{a} + \vec{b} = 2\vec{x}$

$\therefore \vec{b} = \vec{x} + \vec{y}$

Let θ be the angle between $\vec{x}$ and $\vec{y}$ then

$\left|\vec{a}\times\vec{b}\right| = \left|\left(\vec{x}-\vec{y}\right)\times\left(\vec{x}+\vec{y}\right)\right|$

$= 2\left|\vec{x}\times\vec{y}\right|$

$\Rightarrow 2\left|\vec{x}\right|\left|\vec{y}\right|\sin\theta = 2\sqrt{\left|\vec{x}\right|^2\left|\vec{y}\right|^2\sin^2\theta}$

$= 2\sqrt{\left|\vec{x}\right|^2\left|\vec{y}\right|^2\left(1-\cos^2\theta\right)}$

$= 2\left\{4.4-\left(\vec{x}.\vec{y}\right)^2\right\}^{1/2}$

$= 2\left\{2^{2^2}-\left(\vec{x}.\vec{y}\right)^2\right\}^{1/2}$

$\therefore k = 2$

66. A

Sol. Given $y^2 + \frac{h}{2\cos\theta}xy + \frac{3}{4}\tan\theta x^2 = 0$

Let two lines be $y = m_1 x$ and $y = m_2 x$.

$\therefore m_1 + m_2 = \dfrac{-1}{2\cos\theta}$; $m_1 m_2 = \dfrac{3}{4}\tan\theta$

$\therefore m_1, m_2$ are two roots of

$m^2 + \dfrac{h}{2\cos\theta} m + \dfrac{3}{4}\tan\theta = 0$

$\Rightarrow 4\cos\theta\; m^2 + 2hm + 3\sin\theta = 0 \qquad -----(1)$

As one of the lines bisects the coordinate axes,

$m = 1 \text{ or } m = -1$

$\therefore$ Putting $m = \pm 1$ in $equation(1)$ we get

$4\cos\theta + 3\sin\theta = \pm 2h$

This relation holds good if $-5 \le 2h \le 5$

$\Rightarrow \dfrac{-5}{2} \le h \le \dfrac{5}{2}$

$\therefore h = -2, -1, 0, 1, 2$ are 5 integral values of h

67. A

Sol. Since $0 < n+1 < 2n+1 \qquad (as\ n \to \infty)$

$\therefore n^2 < n^2 + n + 1 < n^2 + 2n + 1$

$\Rightarrow n < \sqrt{n^2 + n + 1} < (n+1)$

$\Rightarrow \left[\sqrt{n^2 + n + 1}\right] = n$

$\therefore \lim\limits_{n\to\infty} \dfrac{1+2+3+.............+n}{n^2} = \dfrac{1}{2}$

68. B

Sol. The given expression is the coefficient of x^4 in

${}^4C_0 (1+x)^{404} - {}^4C_1 (1+x)^{303} + {}^4C_2 (1+x)^{202} - {}^4C_3 (1+x)^{101} + {}^4C_4$

$= \left((1+x)^{101} - 1\right)^4$

$= \left({}^{101}C_1 x + {}^{101}C_2 x^2 + ...\right)^4$

$= (101)^4$

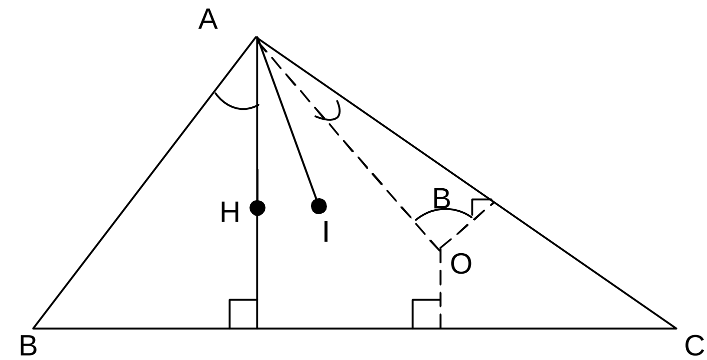
69. A

Sol. $\vec{r} = \dfrac{\left(\vec{r}.(\hat{n}_2 \times \hat{n}_3)\right)\hat{n}_1 + \left(\vec{r}.(\hat{n}_3 \times \hat{n}_1)\right)\hat{n}_2 + \left(\vec{r}.(\hat{n}_1 \times \hat{n}_2)\right)\hat{n}_3}{[\hat{n}_1, \hat{n}_2, \hat{n}_3]}$

$= \dfrac{\hat{n}_1 + 2\hat{n}_2 + 3\hat{n}_3}{[\hat{n}_1, \hat{n}_2, \hat{n}_3]}$

70. D

Sol.



Since, $\angle BAH = 90^0 - B = \angle OAC$

$\angle BAI = A/2 = \angle IAC$

$\therefore \angle HAI = \frac{A}{2} - \left(90^0 - B\right) = \angle IAO$

$\Rightarrow \alpha = \beta$

71. A

Sol. $A_k A_{k+1} = \begin{bmatrix} C_{k+1} & 0 \\ 0 & C_k \end{bmatrix} \begin{bmatrix} C_k & 0 \\ 0 & C_{k+1} \end{bmatrix}$

$= \begin{bmatrix} C_{k+1}C_k & 0 \\ 0 & C_k C_{k+1} \end{bmatrix}$

$\therefore A_1A_2 + A_2A_3 + ... + A_{n-1}A_n = \begin{bmatrix} C_0C_1 + C_1C_2 + ...+ & 0 \\ 0 & C_1C_2 + C_2C_3 + ... + C_nC_{n+1} \end{bmatrix}$

$\therefore a = C_0C_1 + C_1C_2 + ... + C_{n-1}C_n$

$= {}^{2n}C_{n-1}$

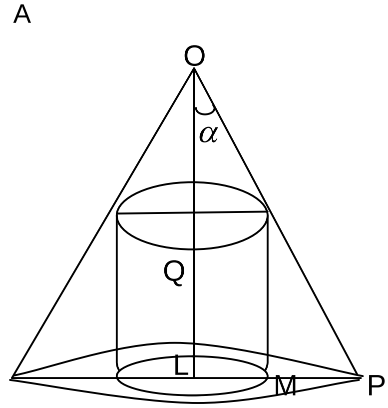
& $b = C_1C_2 + ... + C_nC_{n+1}$

$= {}^{2n}C_{n-1}$

$\therefore a = b$

72. A

Sol.



Let H be the height of the cone and α be its semi vertical angle. Suppose that x is the radius of the inscribed cylinder and h be its height.

$\therefore\ h = QL = OL - OR$

$= H - x\cot\alpha$

V = vol. Of the cylinder

$= \pi x^2 \left(H - x\cot\alpha\right)$

Also, $p = \frac{1}{3}\pi\left(H\tan\alpha\right)^2 H$

$\frac{dV}{dx} = \pi\left(2Hx - 3x^2\cot\alpha\right)$

$\therefore \frac{dV}{dx} = 0 \Rightarrow x = 3, \quad x = \frac{2H}{3}\tan\alpha$

$\therefore \left(\frac{d^2V}{dx^2}\right)_{x=\frac{2H}{3}\tan\alpha} = -2\pi H < 0$

$\therefore$ V is maximum when $x = \frac{2H}{3}\tan\propto$

And $q = V_{max} = \pi\frac{4}{9}\tan^2\alpha H^2\left(\frac{1}{3}H\right)$

$= \frac{4}{9}p$

Hence $p : q = 9 : 4$

73. C

Sol. $f\left(x\right) = \min\left\{\left|\tan x\right|, \left|\cot x\right|\right\}$

$\therefore$ Required area $= 2\int\limits_0^{\pi/4} \tan x\ dx + 2\int\limits_{\pi/4}^{\pi/2} \cot x\, dx$

$= 2\left[\left|\ell n\left|\sec x\right|\right|_0^{\pi/4} + \left|\ell n\right|\sin x\left|\right|_{\pi/4}^{\pi/2}\right]$

$= 2\left[\ell n\sqrt{2} - \ell n1 + \ell n1 - \ell n\frac{1}{\sqrt{2}}\right]$

$= 2\left[2\ell n\sqrt{2}\right]$

$= \ell n4$

74. D

Sol. Numbers are of the form : $7K,\ 7K+1, 7K+2, 7K+3, 7K+4, 7K+5, 7K+6$ their squares : $7K,\ 7K+1, 7K+4, 7K+2, 7K+2, 7K+1, 7K+1$. So, for $a^2+b^2+c^2$ to be a multiple of 7, either all the three squares should be of the form $7K$ or they belong two the categories $7K+1, 7K+2, 7K+4$ separately.

$\therefore$ Required probability $= \left(\frac{1}{7}\right)^3 + 3!\left(\frac{2}{7}\right)\left(\frac{2}{7}\right)\left(\frac{2}{7}\right) = \frac{1}{7}$

75. C

Sol. Shortest distance $= \dfrac{1}{\sqrt{78}}$

$\therefore \quad \dfrac{1}{\text{shortest distance}} = \sqrt{78}$

76. A

Sol. Since ellipse and hyperbola intersect orthogonally, they are confocal hence $a = 2$

Let $P\left(x_1, y_1\right)$ be the point of intersection of both the curves therefore,

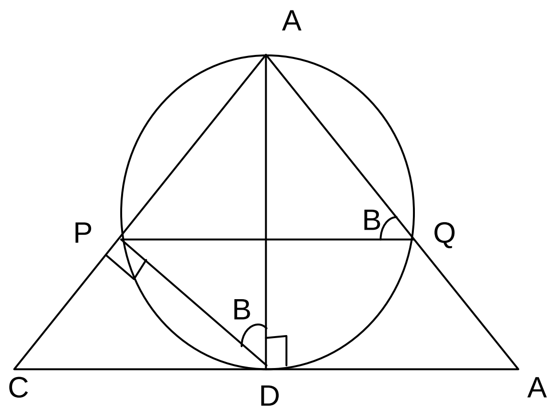
$4x_1^2 + 9y_1^2 = 36$ and $4x_1^2 - y_1^2 = 4$

$\Rightarrow x_1^2 + y_1^2 = 5$

Hence the equation of circle is $x^2 + y^2 = 5$

77. C

Sol.



In ΔAPQ, $\dfrac{PQ}{\sin A} = \dfrac{AP}{\sin \angle AQP}$

$\Rightarrow \dfrac{PQ}{\sin A} = \dfrac{AP}{\sin \angle ADP} \qquad \left(\because \angle AQP = \angle ADP\right)$

$\Rightarrow \dfrac{PQ}{\sin A} = \dfrac{AP}{\sin B}$

$\Rightarrow \dfrac{PQ}{\sin A} = \dfrac{AD \sin B}{\sin B} \qquad \left(\because \sin B = \dfrac{AP}{AD}\right)$

$\Rightarrow PQ = AD \sin A \qquad ---(1)$

In $\Delta\, ADC$, $\sin C = \dfrac{AD}{AC} \;\Rightarrow AD = b \sin C \qquad ----(2)$

$\therefore$ from (1), $PQ = b \sin C \sin A$

$= 2R \sin A \sin B \sin C$

$PQ = \dfrac{\Delta}{R}$

78. D

Sol. The function for $x > 2$ can be redefined as,

$$f(x) = \int_0^1 \left(1+(1-t)\right)dt + \int_1^x \left(1+(t-1)\right)dt$$

$$= 2 - \frac{1}{2} + \frac{x^2}{2} - \frac{1}{2}$$

$$= \frac{x^2}{2} + 1$$

$$RHL = \lim_{h\to 0^+}\left\{\frac{(2+h)^2}{2}+1\right\} = 3$$

$$LHL = \lim_{h\to 0^-}\left\{5(2-h)-7\right\} = 3$$

$$\therefore LHD \neq RHD$$

79. C

Sol.

p	q	$(\sim q \wedge p) \vee (p \vee \sim q)$
T	T	T
T	F	T
F	T	T
F	F	T

80. B

Sol. In I_1 putting $x = \dfrac{4y}{3+y}$ gives

$$I_1 = \int_0^1 \left(\frac{4y}{3+y}\right)^{5/2} \left(\frac{3(1-y)}{3+y}\right)^{7/2} \frac{dy}{(3+y)^2}$$

$$= 864\sqrt{3}\int_0^1 \frac{x^{5/2}(1-x)^{7/2}}{(3+x)^8}dx = 864\sqrt{3}\, I_2$$

SECTION – B

81. 00008.00

Sol. E_1: Dot removed from odd face.

$$P(E_1) = \frac{9}{21}$$

E_2: Dot removed from even face.

$$P(E_2) = \frac{12}{21}$$

E: Die shows odd numbers of dots.

$$P(E) = P(E \cap E_1) + P(E \cap E_2) = \frac{11}{21}$$

82. 00009.00

Sol. $\left|Z\left(a+bi\right)-Z\right|=\left|(a+bi)Z\right|$

$\Rightarrow a=\dfrac{1}{2}$

$a^2+b^2=64$

83. 00002.00

Sol. $\begin{bmatrix} 1 & 2\text{n} & \text{na}+8\sum_{k=0}^{n-1} k \\ 0 & 1 & 4\text{n} \\ 0 & 0 & 1 \end{bmatrix}$

84. 00009.00

Sol. A.M.=G.M.

$\dfrac{\text{r}_1}{2}=\dfrac{\text{r}_2}{4}=\dfrac{\text{r}_3}{5}=\dfrac{\text{r}_4}{8}=\text{k}$

85. 00008.00

Sol. $\text{t}_1+\text{t}_2=2$

$\dfrac{2-\text{m}_2}{1+2\text{m}_2}=\sqrt{3}$

and $\dfrac{\text{m}_3-2}{1+2\text{m}_3}=\sqrt{3}$

$\text{t}_1+t_2+t_3=\dfrac{p}{q}=\dfrac{3}{11}$

86. 00007.00

Sol. Clearly a team of 4 with two batsman and two bowler can be formed as

Case-I: 2 bowler+2batsman

Number of ways= ${}^4\text{C}_2\times{}^4\text{C}_2=36$

Case-II: 2 bowler+1 batsman +1 all-rounder

${}^4\text{C}_2\times{}^4\text{C}_1\times{}^1\text{C}_1=24$

Case-III: 1bowler +2 batsman +1 all-rounder

${}^4\text{C}_1\times{}^4\text{C}_2\times{}^1\text{C}_1=24$

$\therefore$ Probability that all-rounder is selected= $\dfrac{24+24}{36+24+24}=\dfrac{4}{7}$

$\therefore \quad p=7$

87. 00005.00

88. 00000.32

89. 00031.00

Sol. $$\frac{(x-3)^{\frac{-|x|}{x}}\sqrt{(x-4)^2}\,(17-x)}{\sqrt{-x}\left(-x^2+x-1\right)\left(|x|-32\right)}<0$$

For $\sqrt{-x}$ to be defined $x<0$

$$\Rightarrow \frac{-|x|}{x}=1$$

Since $\sqrt{(x-4)^2}, \sqrt{-x}$ are positive & $-x^2+x-1$ is always negative

$$\frac{(x-3)(x-17)}{x+32}>0$$

$x<0$

$$\Rightarrow \frac{1}{x+32}>0 \Rightarrow x=31$$

90. 00833.33

Sol. $$\because \int_a^x e^{f(t)}dt=\int_0^x g(x-t)dt+2x+3$$

$$\Rightarrow \int_a^x e^{f(t)}dt=\int_0^x g(t)dt+2x+3$$ (Using King Property)

Differential both sides, we get

$e^{f(x)}=g(x)+2$

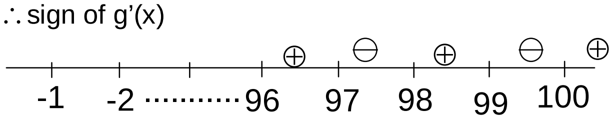
$\Rightarrow g(x)=e^{f(x)}-2$

$\Rightarrow g'(x)=e^{f(x)}.f'(x)$

$\because e^{f(x)}$ is always greater than zero.

$\therefore$ sign of g'(x) is same as sign of $f'(x)$.

$\therefore$ sign of g'(x)



Clearly, local extremum (maximum or minimum) will occur at x=99,97,95,….,3,1

$\therefore$ Sum of all the values=1+3+5+…+99= $\frac{50}{2}[2\times1+(50-1)\times2]=2500$